

Tsomakian Konstantinos Exercise 1

Principal Component Analysis

The first thing that I have to do is to build a function that computes the principal component analysis algorithm based on the eigen value decomposition approach and one based on the Singular Value Decomposition approach. The goal in both approaches is the same. I have to find a set of vectors that point to the direction of the highest variance paired with a set of values that indicate the magnitude of the variance. In the first approach the vectors are the eigen vectors of the covariance matrix and the values the corresponding eigen values, whereas in the SVD approach the vectors of the V matrix (of the decomposed X matrix) and the values are the singular values.

Regarding the eigen value decomposition approach:

- Center the data. This step involves subtracting the mean from each observation.
- Compute the covariance matrix of the centered data.
- Compute the eigen values and the eigen vectors of the covariance matrix
- Project the data to the eigen vectors that correspond to the highest eigen values.

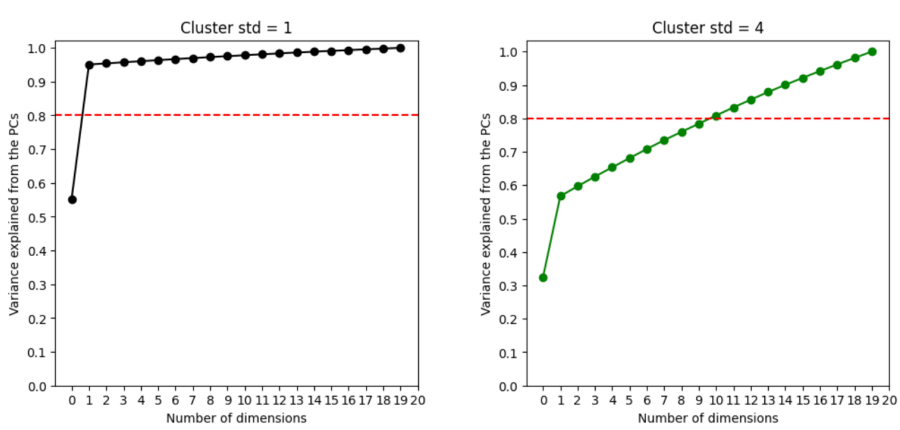
Regarding the SVD approach:

- Apply an SVD decomposition in the centered dataset
- The columns of the V matrix correspond to the principal components and the diagonal of the Sigma matrix are the singular values
- Compute the square of the singular values to get the eigen values

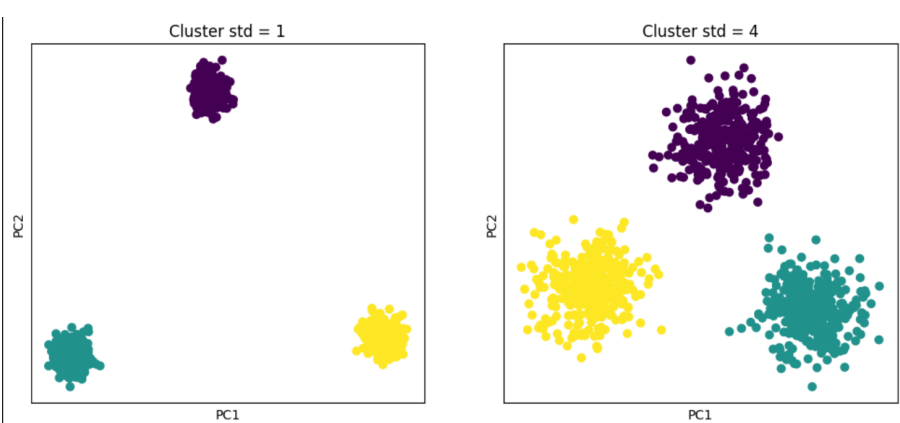
Finally I have to build the function where the user specifies the desired approach for PCA:

Questions

***How many dimensions must we keep to maintain 80% of the total variance?
Show the relevant plot (e.g., variance explained as a function of dimensions).
Project the data by keeping these dimensions.***

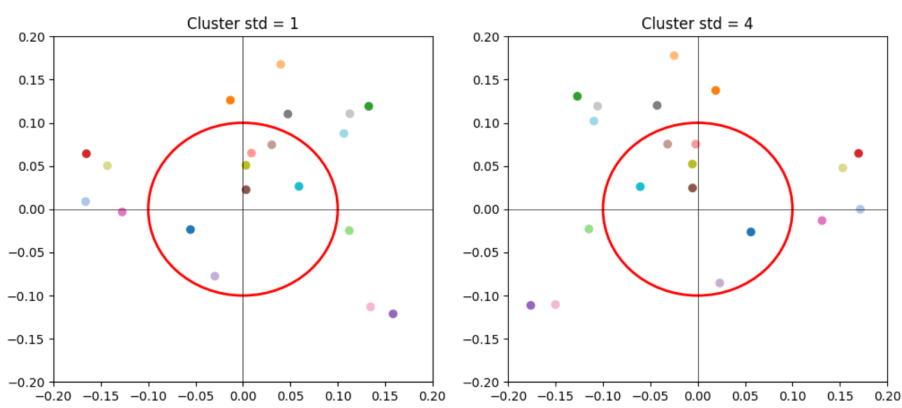


The plot on the left describes the cumulative explained variance against the number of dimensions. To explain 80 percent of the variance 2 dimensions (first 2 PCs) are needed for the dataset with cluster std equal to 1. On the other hand 11 dimensions are needed to explain the same amount of variance on the dataset with cluster std equal to 4.



Here the the centered data are projected to the first two principal components and the resulting data are plotted. As expected the data points of the first dataset form more compact clusters compared to the datapoints of the second cluster.

Plot the loadings of the first two principal components in the same plot. Do you think that some features are not needed? If yes, explain why. If no, explain why.



Loadings represent the weight that each feature holds in forming each principal component. As a result loadings with values closer to zero are of less importance in forming the principal components. So when plotting the loadings of the first two principal components datapoints closer to the origin of the coordinate system represent features of low importance. Here a circle with radius of 0.1 has been arbitrarily chosen to identify features of low importance.